Retail Sales Analysis Using SQL

This project analyzes retail sales data using PostgreSQL to derive business insights related to customer behavior, product categories, sales performance, and purchasing trends. The project demonstrates SQL skills such as filtering, aggregation, grouping, window functions, ranking, and date/time analysis.

Objectives

- Analyze sales performance across product categories.
- Identify top customers based on revenue generation.
- Understand customer demographics and purchasing patterns.
- Determine peak sales periods and best-performing months.
- Generate actionable business insights from transactional data.

Dataset Information

The dataset contains retail transaction records with the following attributes:

Column Name	Description
transactions_id	Unique transaction identifier
sale_date	Date of sale
sale_time	Time of sale
customer_id	Unique customer identifier
gender	Customer gender
age	Customer age
category	Product category
quantity	Quantity purchased
price_per_unit	Price per unit
cogs	Cost of goods sold
total_sale	Total transaction value

SQL Concepts Used

- SELECT Statements
- WHERE Clause
- GROUP BY
- ORDER BY
- Aggregate Functions
- Date Functions
- Window Functions
- Ranking Functions
- CASE Statements
- Subqueries

Business Questions Solved

1. Sales on a Specific Date

Retrieved all transactions made on a particular date.

2. Category-Based Sales Analysis

Identified Clothing category purchases with quantity greater than or equal to 3 during November 2022.

3. Category Revenue Analysis

Calculated total sales generated by each product category.

4. Customer Demographic Analysis

Computed the average age of customers purchasing Beauty products.

5. High-Value Transactions

Identified transactions where total sales exceeded 1000.

6. Gender-wise Category Analysis

Analyzed transaction counts by gender across product categories.

7. Best Performing Month Analysis

Used window functions and ranking to identify the highest-performing sales months.

8. Top Customers

Ranked customers based on total sales generated.

9. Unique Customer Analysis

Determined the number of unique customers purchasing from each category.

10. Sales Shift Analysis

Classified transactions into:

- Morning (≤ 12 PM)
- Afternoon (12 PM – 5 PM)
- Evening (> 5 PM)

Key Insights

- Identified top revenue-generating categories.
- Found the highest-value customers.
- Analyzed customer demographics across product categories.
- Discovered peak sales periods and purchasing trends.
- Evaluated sales distribution across different times of the day.

Tools Used

- PostgreSQL
- SQL
- GitHub

Skills Demonstrated

- Data Cleaning
- Data Exploration
- Business Analysis
- SQL Query Optimization
- Data Aggregation
- Window Functions
- Analytical Thinking

Project Structure

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Retail-Sales-Analysis/
│
├── Retail_Sales_Analysis.sql
├── Dataset.csv
├── README.md
└── Project_Screenshots/
```